Projection-Free Spectral Factorization of the Regularized Determinant

Liang Wang

July 2026

Abstract

The direct moving-cloud Riesz projectors of RH-232 are severely ill-conditioned, but the regularized determinant depends on eigenvalues, not eigenvectors. Let $A \in \mathcal{S}_2$ have nonzero eigenvalues (λ_j) with algebraic multiplicity. For any finite submultiset C, the canonical product splits exactly:

$$\det_2(I - zA) = \prod_{\lambda \in C}(1 - z\lambda)e^{z\lambda} \prod_{\lambda_j \notin C}(1 - z\lambda_j)e^{z\lambda_j}.$$

Both factors are entire and no projector norm appears.

We apply this identity to the selected and resolved omitted roots of every RH-222 endpoint. On a 192-point unit-disk grid, all 6,144 factorization cases pass with maximum absolute discrepancy 1.83×10^{-15}, despite the inherited projector norm 2.26×10^{12}. This establishes a projection-free finite cloud factor. It does not prove that the selected multisets form the correct small-noise divisor or that the complementary products have a locally uniform limit.

1 Why the determinant can bypass the projector

For $A \in \mathcal{S}_2$, the two-regularized determinant is the genus-one canonical product associated with the compact spectrum [1, 2]. Its value is similarity invariant and contains no eigenvector condition number. This distinguishes factor evaluation from invariant-subspace perturbation.

RH-80 formulated the moving factor using a reducing projection because that is the natural route to an operator complement [5]. RH-232 and RH-233 show why the projector norm is a difficult small-noise object [3, 4]. The present paper separates the algebraic factor from that operator realization.

2 Canonical-product factorization

Theorem 2.1 (Finite spectral submultiset factor)**.** *Let $A \in \mathcal{S}_2(H)$ and let (λ_j) denote its nonzero eigenvalues, with algebraic multiplicity. Let C be any finite submultiset. Then*

$$D_A(z) := \det_2(I - zA) = C_A(z)R_A(z),$$

where

$$C_A(z) = \prod_{\lambda \in C}(1 - z\lambda)e^{z\lambda}, \qquad R_A(z) = \prod_{\lambda_j \notin C}(1 - z\lambda_j)e^{z\lambda_j}.$$

Both products define entire functions. The factorization depends only on the chosen eigenvalue multiset and not on any spectral projector norm.

Proof. The eigenvalue sequence belongs to ℓ^2 by the Weyl inequality for Hilbert–Schmidt operators. Hence the genus-one product converges locally uniformly. Removing finitely many factors preserves convergence, and reordering a normally convergent canonical product does not change its value. Multiplying the finite and complementary products recovers the full product. □

Proposition 2.2 (Trace germ)**.** *On the disk of convergence of the logarithmic germ,*

$$\log R_A(z) = -\sum_{n\geq 2} \frac{z^n}{n}\tau_n, \qquad \tau_n = \sum_{\lambda_j \notin C} \lambda_j^n.$$

If A^n is trace class, τ_n equals the trace of A^n after subtracting the selected eigenvalue powers.

Proof. Use $\log((1-w)e^w) = -\sum_{n\geq 2} w^n/n$ and interchange normally convergent sums near the origin. Lidskii's theorem identifies trace-class power traces with algebraic eigenvalue sums [2]. □

Proposition 2.2 identifies the replacement for a failed ideal norm estimate: directly control the complement power traces.

Proposition 2.3 (Selected divisor and logarithmic derivative)**.** *Assume the selected multiset contains no zero values. The zeros of C_A are precisely the reciprocals λ^{-1} for $\lambda \in C$, with their selected algebraic multiplicities, and*

$$\frac{C_A'(z)}{C_A(z)} = -\sum_{\lambda \in C} \frac{z\lambda^2}{1-z\lambda}$$

away from those zeros. Hence the projection-free quotient removes exactly the selected finite divisor and no additional finite zeros.

Proof. The exponential multiplier in each genus-one factor is nonvanishing, so its only zero is $z = \lambda^{-1}$. Differentiating $\log((1-z\lambda)e^{z\lambda})$ gives $-\lambda/(1-z\lambda) + \lambda = -z\lambda^2/(1-z\lambda)$. □

This divisor statement is stronger than a grid identity. Once a finite cloud multiset has been chosen, its determinant factor and multiplicities are defined algebraically even when the corresponding invariant subspace is almost defective.

3 Finite-factor stability without eigenvectors

The absence of projector norms does not make root approximation irrelevant, but it gives the appropriate conditioning scale. Let $f(w) = \log(1-w) + w$, normalized by $f(0) = 0$. If paired finite multisets $C = \{\lambda_j\}_{j=1}^k$ and $\widetilde{C} = \{\widetilde{\lambda}_j\}_{j=1}^k$ satisfy

$$|z\lambda_j| \leq r, \qquad |z\widetilde{\lambda}_j| \leq r < 1 \quad (|z| \leq R),$$

then the normalized logarithms obey

$$\left|\log C_A(z) - \log \widetilde{C}_A(z)\right| \leq \frac{rR}{1-r} \sum_{j=1}^k |\lambda_j - \widetilde{\lambda}_j|. \tag{1}$$

Indeed $f'(w) = -w/(1-w)$ has modulus at most $r/(1-r)$ on the convex disk $|w| \leq r$, and one applies the mean-value integral to every paired factor.

Estimate (1) depends on root errors and distance from the reciprocal divisor, but not on left/right eigenvector angles. It therefore explains why the 10^{12} projector norm and the 10^{-15} product identity can coexist without contradiction.

4 Finite resolved audit

For each endpoint, the candidate bulk multiset is split into the selected shell-complete cloud and the remaining resolved candidate roots. We evaluate all factors at 48 angles on each of the radii 0.25, 0.5, 0.75, and 1.

Quantity	value
Endpoint multisets	32
Grid points per endpoint	192
Factorization cases	6,144
Maximum cloud matching error	0
Maximum product discrepancy	1.8311×10^{-15}
Inherited maximum projector norm	2.2610×10^{12}

Table 1: Projection-free finite product audit.

The computation is not a surprise algebraically; its role is architectural. It verifies that the numerical determinant factor does not inherit the Riesz conditioning catastrophe.

The audit also separates three finite errors that are sometimes conflated: the cloud-matching error, evaluation roundoff in the selected product, and evaluation roundoff in the complementary product. The first is exactly zero for the archived multisets because the same stored roots are partitioned. The quoted 1.83×10^{-15} maximum is therefore an evaluation identity, not an eigenvalue-certification radius.

5 Factorization and the small-noise limit

At fixed noise and fixed finite dimension, Theorem 2.1 settles the algebra. Passing to $\sigma \to 0$ is a different operation because both the ambient matrix and the selected cloud rank change. A valid limiting argument must establish all three of the following:

1. local uniform control of the complementary products;
2. compact-set stability of the selected reciprocal divisor;
3. compatibility of the chosen finite factor with the intended deterministic normalization.

None follows from pointwise finite product multiplication. The first is translated by Proposition 2.2 into an all-order trace problem; the third becomes the coefficient-anchor problem isolated later in the batch.

6 Boundary and next target

Selecting a finite multiset at each positive noise is easy once its roots are listed. The hard questions are now:

1. whether the selected multisets obey a canonical small-noise rule;
2. whether the complementary canonical products form a normal family;
3. whether their coefficient germ is the intended deterministic numerator rather than an over-renormalized remainder.

RH-235 shows that the divergent Hilbert–Schmidt budget does not decide the second question. RH-236 therefore computes the trace germ through order twelve. No Gate A closure or arithmetic zero identification is claimed.

References

[1] Israel Gohberg and Mark Krein. *Introduction to the Theory of Linear Nonselfadjoint Operators*. American Mathematical Society, 1969.

[2] Barry Simon. *Trace Ideals and Their Applications*. American Mathematical Society, second edition, 2005.

[3] Liang Wang. Biorthogonal riesz-cloud projection and its conditioning wall. 2026. RH-232 preprint.

[4] Liang Wang. Why radial shell gaps do not control nonnormal riesz projections. 2026. RH-233 preprint.

[5] Liang Wang. Moving-cloud renormalization and the relative determinant gate. 2026. RH-80 preprint.